

DC CIRCUITS

OHM'S LAW

→ Acc to Ohm's law, P.D across any 2 points of conductor, will be directly proportional to current flow through it.

$$V \propto I$$

$$V = IR, R \rightarrow \text{Resistance (ohm} - \Omega)$$

KIRCHOFF'S LAW : Based on consⁿ of energy and charges.

KVL

KCL

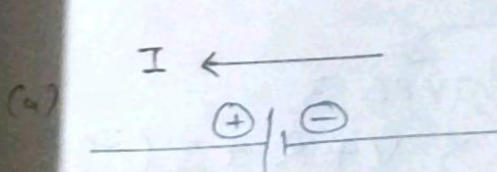
→ deals with consⁿ of energy.

* KIRCHOFF'S VOLTAGE LAW (KVL)

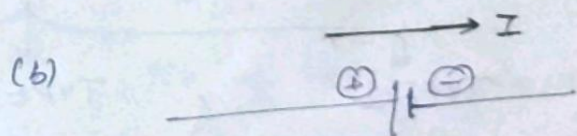
→ Acc to KVL, 'In any closed circuit OR mesh, The Algebraic sum of all EMF's and Voltage drop will be zero.

$$\sum V = \sum \text{EMF} + \sum V.D = 0$$

→ Energy consⁿ, Net energy supplied to circuit = Net energy Consumed by circuit.



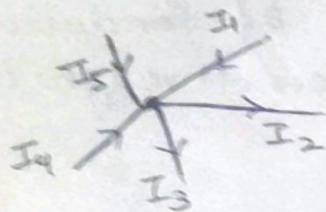
→ Voltage Rise.
When $I \rightarrow \text{L.P to H.P}$
 $\{ \Delta V = +ve \}$



→ Voltage Drop
When $I \rightarrow \text{H.P to L.P}$
 $\{ \Delta V = -ve \}$

* KIRCHOFF'S CURRENT LAW (KCL)

→ Acc to KCL, 'Algebraic sum of all currents meeting at a point or a junction will be zero'.

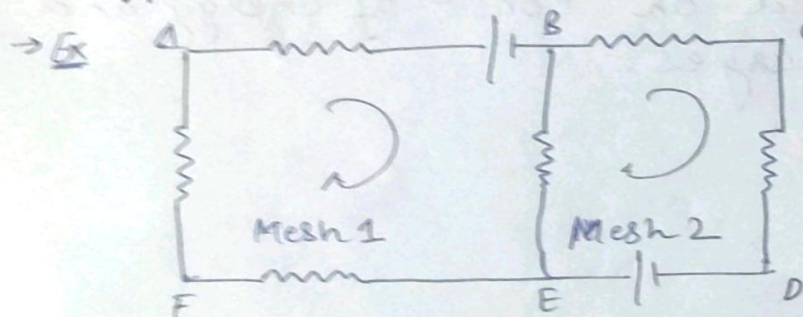


$$\boxed{\sum I = 0}$$

$$\Rightarrow I_1 - I_2 - I_3 + I_4 + I_5 = 0$$

$$I_1 + I_4 + I_5 = I_2 + I_3$$

→ Difference b/w Mesh and A Loop.



Mesh → No of mesh = 2

→ AB EF or BC DE

Loops = 3

→ AB EF, BC DE, AC DE

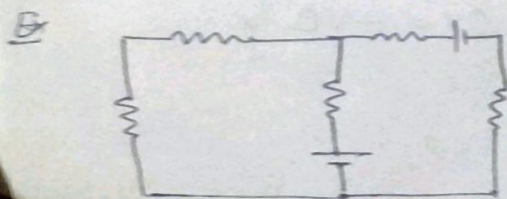
* MESH: It is closed path across which current flows but this closed path shouldn't contain any other closed path.

→ For a loop → also a closed path but can contain another closed path inside itself.

Mesh Analysis Method

TYPE 1

- Voltage source
+
Resistance -



TYPE 2

Voltage source (V)

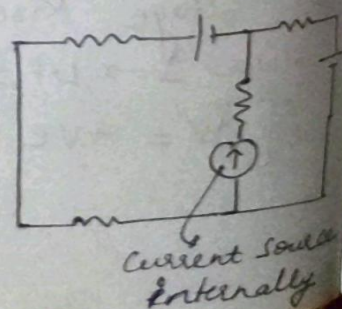
+
Resistance (R)

+
Current sources (I)

↓
But these source are
in outer loop

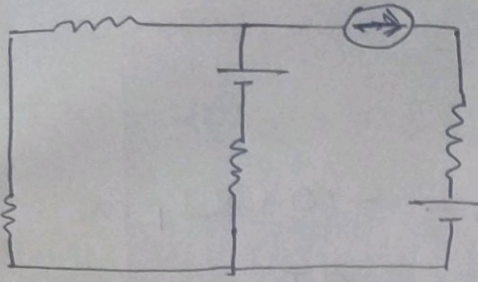
TYPE 3

V + R + I (In inner loop)



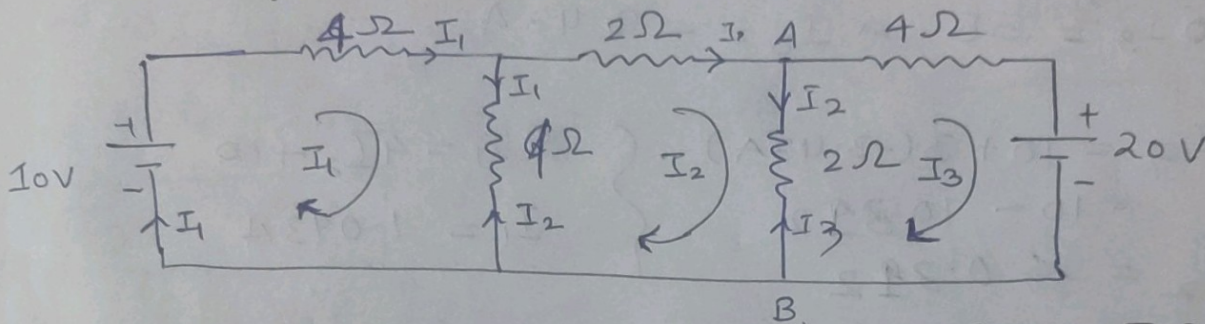
Type 2 circuits.

When current sources are in individual mesh or in external mesh externally.



Q1

Calculate the current in branch AB of 2Ω resistance for given circuit using Mesh analysis



Ans There are 3 meshes.

→ Applying KVL in Mesh 1,

$$+10 - I_1 \times 4 - 4(I_1 - I_2) = 0$$

(Here I_1 is considered +ve)

$$-8I_1 + 4I_2 + 10 = 0 \quad \text{--- (1)}$$

→ Applying KVL in Mesh 2,

$$-2I_2 - 2(I_2 - I_3) - 4(I_2 - I_1) = 0$$

$$-8I_2 + 2I_3 + 4I_1 = 0 \quad \text{--- (2)}$$

→ Applying KVL in Mesh 3,

$$-20 - 4I_3 - 2(I_3 - I_2) = 0 \quad \text{--- (3)}$$

$$-20 - 6I_3 + 2I_2 = 0 \quad \text{--- (3)}$$

STEPS FOR MESH ANALYSIS

1) Firstly, Count the No of Mesh in circuit

2) Decide the sign convention for all mesh

But sign → same for Mesh

3) Now form individual Mesh eqn.

No of eqn = No of Mesh

Solving eqⁿ 1, 2, 3,

$$4I_1 - 8I_2 + 2I_3 = 0$$

$$2I_2 = 20 + 6I_3$$

$$-8I_1 + 4I_2 = -10$$

$$4I_2 = -10 + 8I_1$$

$$\rightarrow 40 + 12I_3 = -10 + 8I_1$$

$$50 + 12I_3 = 8I_1$$

$$25 + 6I_3 - 80 - 24I_3 + 2I_3 = 0$$

$$-46I_3 = 55 \rightarrow I_3 = -3.434$$

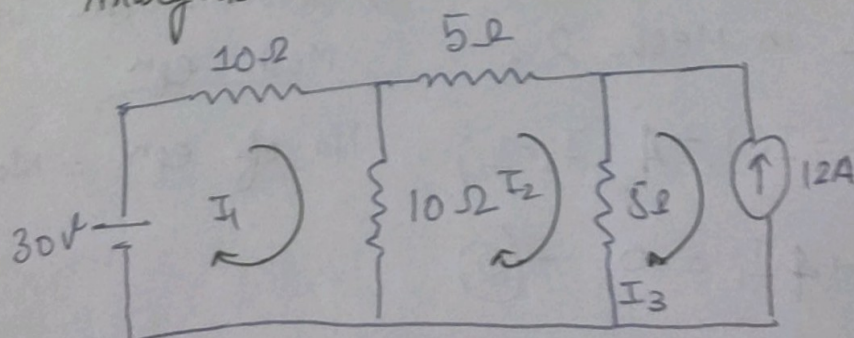
$$\begin{cases} I_2 = 10 + 3(-3.434) \\ \quad = 10 - 10.302 \\ \quad I_2 = -0.302 \end{cases} \quad \begin{cases} 8I_1 = 4I_2 + 10 \\ \quad I_1 = 1.0934 \end{cases}$$

$$I_{AB} = I_2 - I_3$$

$$= -0.302 + 3.434$$

$$I_{AB} = 3.125 \text{ A} \quad \text{Ans}$$

Q2 Type 2. → Find Current in AB Branch using Mesh Analysis.



There are 3 meshes in circuit

Applying KVL in Mesh 1,

$$+30 - 10I_1 - 10(I_1 - I_2) = 0$$

$$+30 - 20I_1 + 10I_2 = 0$$

$$20I_1 - 10I_2 = 30 \Rightarrow 2I_1 - I_2 = 3 \quad \text{--- (1)}$$

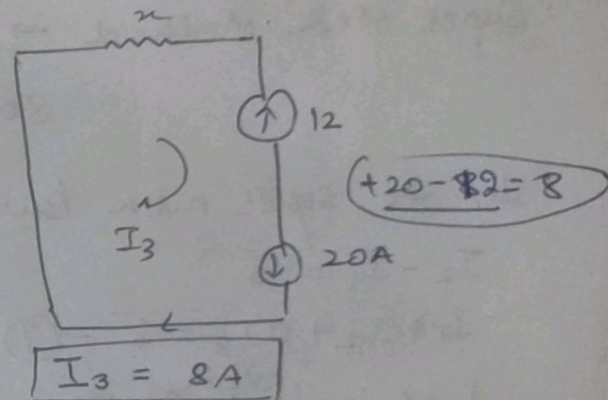
Applying KVL in Mesh 2,

$$-5I_2 - 10(I_2 - I_1) - 5(I_2 - I_3) = 0$$

$$-20I_2 + 10I_1 + 5I_3 = 0$$

$$10I_1 - 20I_2 + 5I_3 = 0 \quad \text{--- (2)}$$

Note:-



$$I_3 = -12A$$

$$10I_1 - 20I_2 - 60 = 0$$

$$I_1 - 2I_2 = 6$$

$$2I_1 - I_2 = 3$$

$$2I_1 - 4I_2 = 12$$

$$\begin{array}{r} - \quad + \quad - \\ \hline +3I_2 = -9 \end{array}$$

$$I_2 = -3A \Rightarrow$$

$$I_1 = 6 - 6$$

$$I_1 = 0A$$

$$I_{AB} = I_2 = -3A$$

Ans

$$V_{AB} = IR$$

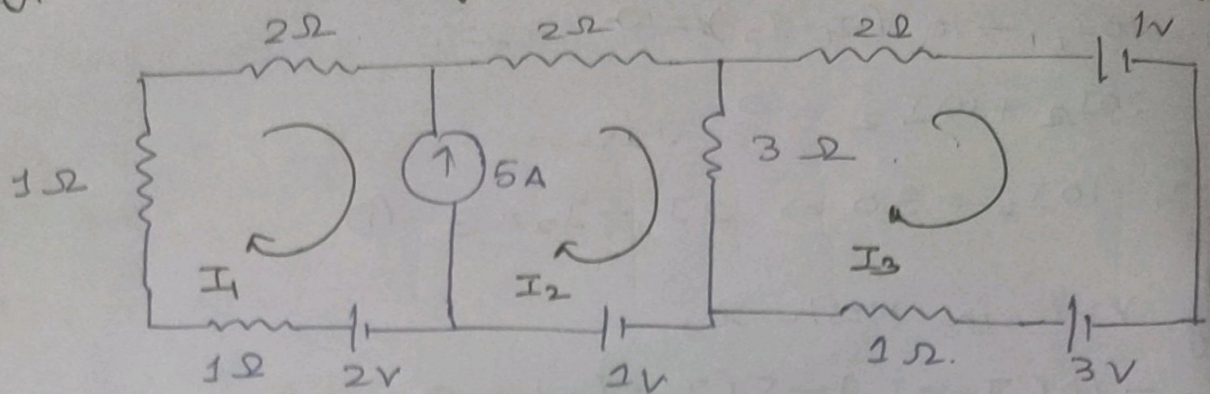
$$V_{AB} = -3 \times 5 = -15V$$

$$P = I^2 R$$

$$P = 9 \times 5$$

$$P = 45W$$

Q3 (Type 3) Find voltage across 3Ω resistor using KVL



→ There are 3 meshes

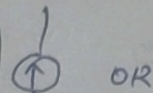
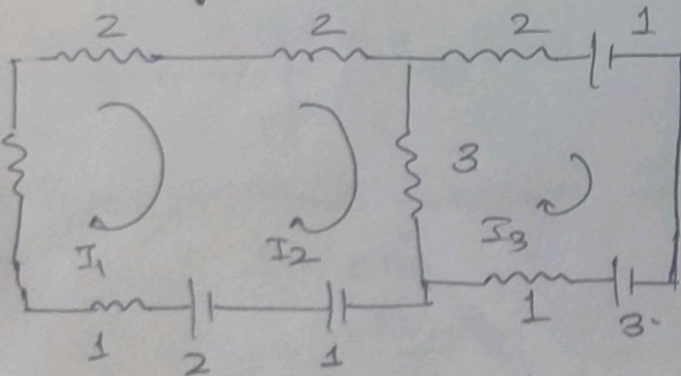
Super Mesh Method → Special method ~~des~~ supported for solving type 3 ~~ex~~ mesh.

Acc to super mesh law,

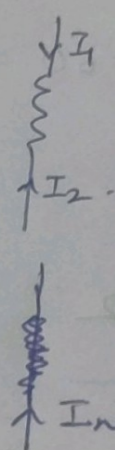
$$I_2 - I_1 = 5$$

$$-I_1 + I_2 + 0I_3 = 5 \quad \text{--- (1)}$$

Applying ~~KVL~~ ^{KVL in} Super Mesh,



OR



dirn of net current is eqd. dirn of bigger current
let I_2 be more current

$$I_{net} = I_2 - I_1$$

$$\Rightarrow +2 - I_1 - I_1 - 2I_1 - 2I_2 - 3(I_2 - I_3) + 1 = 0$$

$$-4I_1 - 5I_2 + 3I_3 + 3 = 0 \quad \text{--- (2)}$$

Applying KVL in Super Mesh 3,

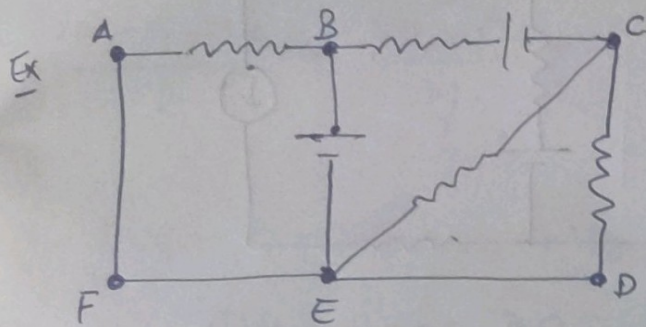
$$-3(I_3 - I_2) - 2I_3 - 1 + 3 - I_3 = 0$$

$$-6I_3 + 3I_2 + 2 = 0 \quad \text{--- (3)}$$

$$I_1 =$$

KIRCHOFF'S CURRENT LAW

- Difference b/w junction and a node.



Junction: It is a pt where current divide.

- ~~nodes~~ It is a pt ^{or any} in given electrical circuit where three or more than three branches combine.

* Node: - It is a pt where two or more than 2 branches combine.

- It is not necessary that current will divide on node. but it can divide.

In above example,

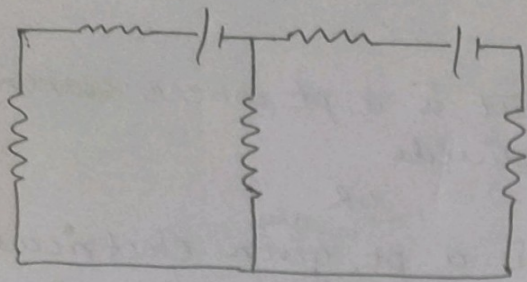
No of Node = 3 (B, C, D)

No of junction = 3 (B, C, D)

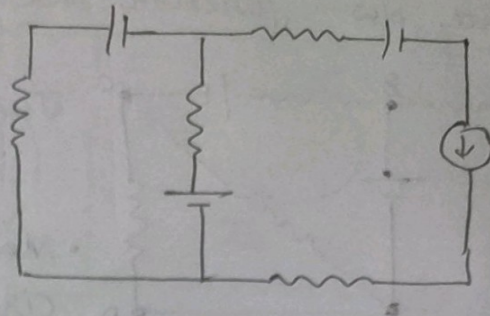
* Steps for solving Numerical using Nodal Analysis:-

1. Identify the principle nodes or junctions present in network
2. Assign a junction potential on each junction w.r.t the assigned reference junction having value $V_0 = 0V$
3. Assuming all the currents in outgoing dirⁿ from each junction from all KCL equation.
4. Solve the equation to calculate value of junction potential
5. Using the individual junction potential find value of req^d electrical quantity.

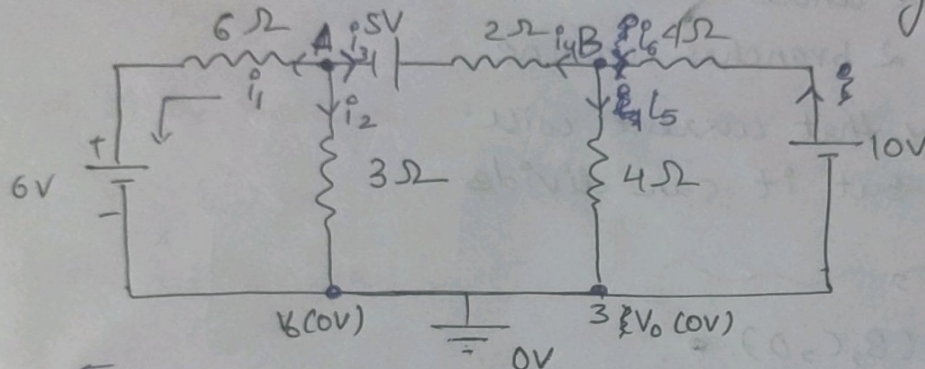
Type-2 (Resistance + voltage)



Type-2 (R+V+Current Source)



Q1 Find the current in 3Ω resistor using KCL



Ans. $\sum I = 0$
Applying KCL at j^{th} (A).

$$I_1 = \frac{V_A - V_0 - 6}{6}, \quad I_2 = \frac{V_A - 0}{3}$$

$$I_1 = \frac{V_A - 6}{6}, \quad I_3 = \frac{V_A + 5 - V_B}{2}$$

$$I_1 + I_2 + I_3 = 0 \quad (\text{KCL eqn})$$

$$\frac{V_A - 6}{6} + \frac{V_A}{3} + \frac{V_A + 5 - V_B}{2} = 0$$

$$V_A - 6 + 2V_A + 3V_A + 15 - 3V_B = 0$$

$$6V_A - 3V_B + 9 = 0$$

$$2V_A - V_B + 3 = 0 \quad \text{--- (1)}$$

Applying KCL at $J^n(B)$.

$$I_4 + I_5 + I_6 = 0$$

$$\frac{V_B - 5 - V_A}{2} + \frac{V_B - 0}{4} + \frac{V_B - 10 - 0}{4} = 0$$

$$2V_B - 10 - 2V_A + V_B + V_B - 10 = 0$$

$$4V_B - 2V_A = 20$$

$$2V_B - V_A = 10 \quad \text{--- (i)}$$

Solving (i) & (ii), we get

$$2V_B - 10 = V_A$$

$$2(2V_B - 10) - V_B + 3 = 0$$

$$4V_B - 20 - V_B + 3 = 0$$

$$3V_B - 17 = 0 \Rightarrow V_B = 17/3 \text{ V}$$

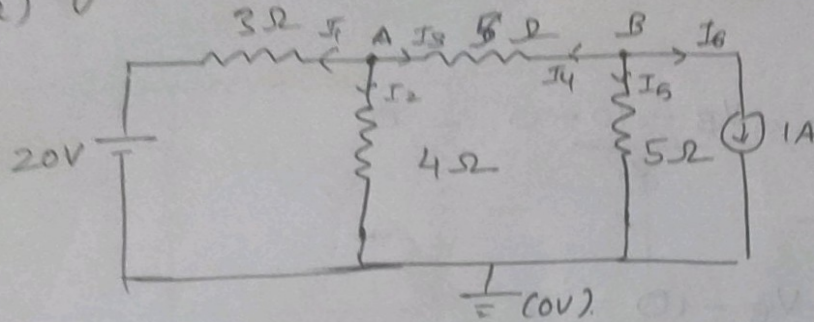
$$V_A = \frac{34}{3} - 10 = \frac{4}{3} \text{ V}$$

$$I_3 = \frac{\frac{4}{3} - \frac{17}{3} + 5}{2} = \frac{\frac{4 - 17 + 15}{3 \times 2}}{2} = \frac{1}{3} \text{ A}$$

$$\boxed{I_3 = 1/3 \text{ A}}$$

$$I_2 = \frac{4}{9} = \underline{\underline{0.44 \text{ A}}}$$

Q2 Using Nodal Analysis, find the current in 5Ω branch.
(Type-2)



⇒ Apply KCL at $j^n(A)$,

$$I_1 + I_2 + I_3 = 0$$

$$\frac{V_A - 20 - 0}{3} + \frac{V_A}{4} + \frac{V_A - V_B}{3} = 0$$

$$\frac{4V_A - 80 + 3V_A + 2V_A - 2V_B}{12} = 0$$

$$\frac{20V_A - 400 + 15V_A + 12V_A - 12V_B}{60} = 0$$

$$4V_A - 2V_B = 80$$

$$47V_A - 12V_B = 400 \quad \text{--- (1)}$$

⇒ Applying KCL at $j^n(B)$,

$$I_4 + I_5 + I_6 = 0$$

$$I_6 = 1A$$

$$\frac{V_B - V_A}{5} + \frac{V_B}{5} + 1 = 0$$

$$V_B - V_A + V_B + 5 = 0$$

$$2V_B - V_A + 5 = 0 \quad \text{--- (2)}$$

Solving (1) & (2)

$$V_A = 9.02V$$

$$V_B = 2.012V$$

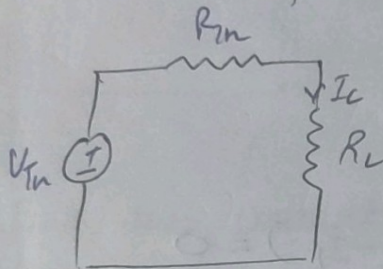
$$I_2 = \frac{V_A}{4} = \frac{9.02}{4} = \underline{\underline{2.24 \text{ A}}}$$

THEVENIN'S THEOREM

→ Acc to thevenin's theorem, 'Any linear bilateral network irrespective of its complexities can be reduced into a thevenin's equivalent circuit having thevenin's ^{open} circuit voltage V_{Th} in series with thevenin equivalent resistance R_{Th} along with load resistance R_L .

* Note:-

- Linear Bilateral Network: It is a circuit or network that exhibit two distinct characteristic behavior.
 - bilateral elements that allow current in both dirn & offer the same impedance in either dirn.



$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

(Thevenin equivalent circuit).

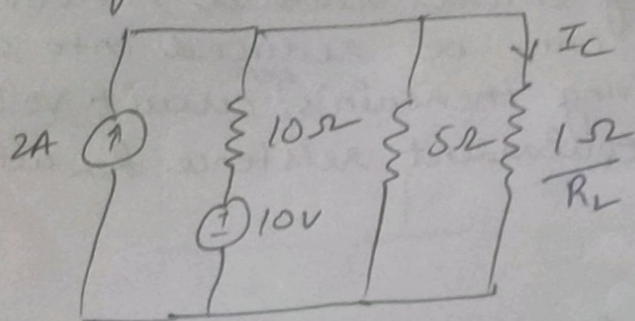
Steps for solving thevenin circuit:

- (I): Identify load resistance R_L .
- (II): Remove load resistor and calculate open circuit potential across two open ends. This will be Thevenin equivalent voltage V_{Th} .
- (III): Again remove the load resistor & replace all active sources by their internal resistance.
- (IV): Calculate equivalent resistances across open ends. This will be thevenin equivalent resistance R_{Th} .
- (V): Draw thevenin equivalent for given network.

Calculate load current I_L by using identity:-

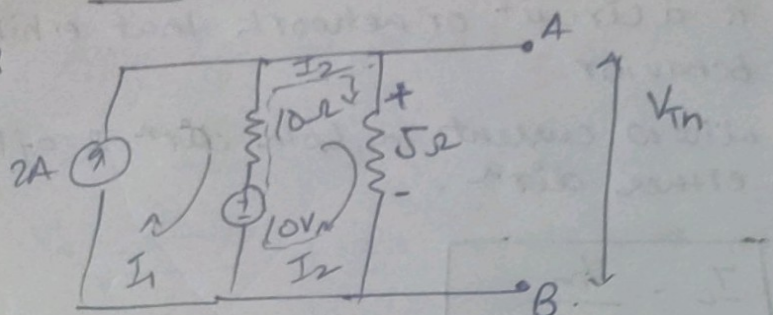
$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

Q1. Find the current through 1Ω resistor in given circuit using theorem.



* Note:-
 R_L is resistance we choose about which current is to find

Ans



* current travel from HP to LP

Applying KVL in mesh 2,

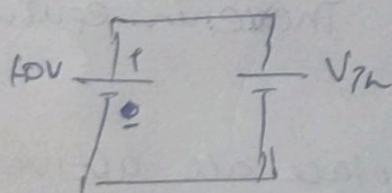
$$-10(I_1 - I_2) - 10 - 5I_2 + 10 - 10(I_2 - I_1) = 0$$

$$-5I_2 + 10 - 10I_2 + 10I_1 = 0$$

$$I_1 = 2A$$

$$10I_1 - 15I_2 + 10 = 0$$

$$20 = 15I_2 \Rightarrow \boxed{I_2 = 2A}$$



$$-V_{th} + 10 = 0$$

$$\boxed{V_{th} = 10V}$$

$$V = IR$$

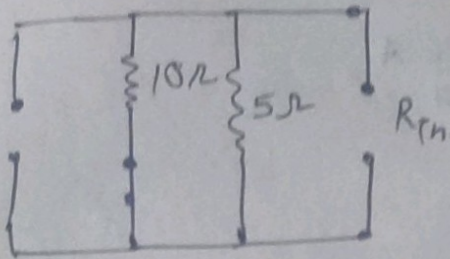
$$V = 10V$$

* Note:-

Active source $\rightarrow$ Voltage and current sources.

Voltage source : $0\Omega (I \cdot R) \rightarrow$ Short circuit

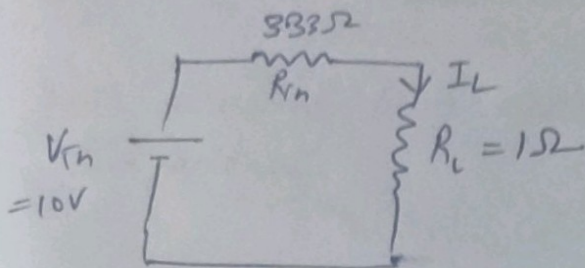
Current source : $\infty (I \cdot R) \rightarrow$ open circuit



Now find equivalent resistance that is R_{Th}

$$R_{Th} = \frac{10 \times 5}{15} = \frac{50}{15} = \frac{10}{3} \Omega$$

$$R_{Th} = 3.33 \Omega$$

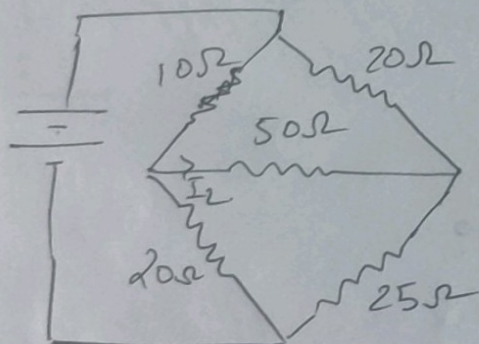


$$I_L = \frac{V_{Th}}{R_L + R_{Th}} = \frac{10}{1 + \frac{10}{3}} = \frac{30}{13}$$

$$I_L = 2.31 A$$

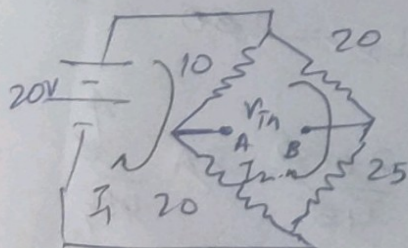
Q2 Find the current through 50Ω resistor using thevenin theorem.

Ans



$$(I) R_L = 50 \Omega$$

(I) Removing R_L to calculate V_{Th}



Applying KVL in Mesh 1,

$$+20 - 10I_1 + 20I_2 - 20I_1 + 20I_2 = 0$$

$$+20 - 30I_1 + 30I_2 = 0$$

$$3I_1 - 3I_2 = 2 \quad \text{--- (1)}$$

Applying KVL in Mesh 2,

$$-20I_2 - 25I_2 - 20(I_2 - I_1) - 10(I_2 - I_1) = 0$$

$$-20I_2 - 25I_2 - 20I_2 + 20I_1 - 10I_2 + 10I_1 = 0$$

$$-45I_2 + 30I_1 = 0 \rightarrow 2I_1 = 3I_2 \quad \text{--- (2)}$$

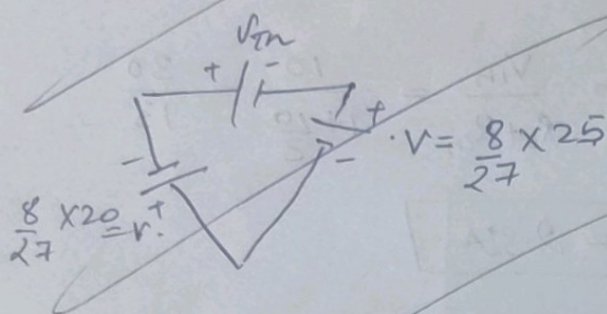
$$20I_1 = 65I_2$$

$$I_1 = \frac{13}{4} I_2 \Rightarrow I_1 = \frac{13}{4} \times \frac{8^2}{27} = \frac{26}{27} A$$

$$8\left(\frac{13}{4} I_2\right) - 3I_2 = 2$$

$$39I_2 - 3I_2 = 8$$

$$27I_2 = 8 \Rightarrow I_2 = \frac{8}{27}$$



$$-V_{th} - \frac{200}{27} - \frac{160}{27} = 0$$

$$-V_{th} = -\frac{360}{27} \Rightarrow V_{th} = \frac{40}{3} V$$

$$-75I_2 + 30I_1 = 0$$

$$30I_1 = 75I_2 \Rightarrow I_1 = \frac{25}{10} I_2$$

$$3\left(\frac{25}{10} I_2\right) + 3I_2 = -2$$

$$-\frac{75}{10} I_2 + 3I_2 = -2 \Rightarrow \frac{45}{10} I_2 = 2$$

$$3I_1 - 3I_2 = 2$$

$$3I_1 = 4$$

$$I_1 = \frac{4}{3} A$$

$$3I_1 - \frac{4}{3} = 2$$

$$9I_1 - 4 = 6$$

$$9I_1 = 10$$

$$I_1 = \frac{10}{9} A$$

$$I_2 = \frac{2000}{45} = 8A$$

$$I_2 - I_1 = \frac{-6}{9}$$

$$V = \frac{10 \times 2}{9 \times 3}$$

$$V_2 = -\frac{40}{3}$$

$$V = \frac{4}{9} \times 25 = \frac{100}{9}$$

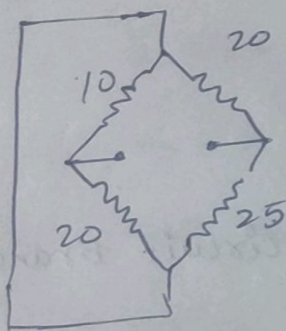
$$\begin{array}{r} 2 \cdot 2 \\ \overline{9 \sqrt{20}} \\ 18 \\ \hline 20 \end{array}$$

$$V_{Th} - \frac{100}{9} + \frac{40}{3} = 0$$

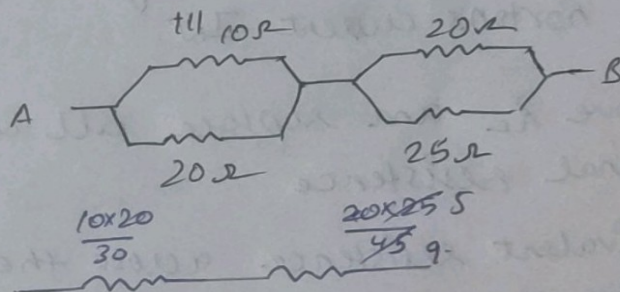
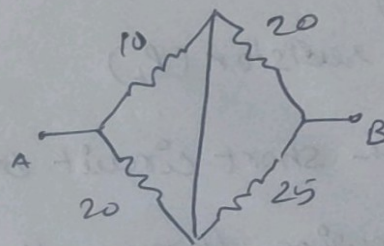
$$V_{Th} = -\frac{20}{9} \Rightarrow V_{Th} = 20/9 \text{ V.}$$

$$V_{Th} = 2.22 \text{ V}$$

Again removing R_L & replacing all current source by their internal Impedence.

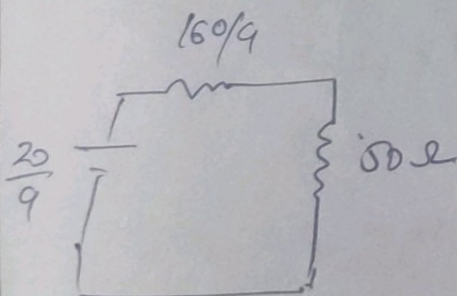


$\equiv$



$$R_{Th} = \frac{20}{3} + \frac{100}{9}$$

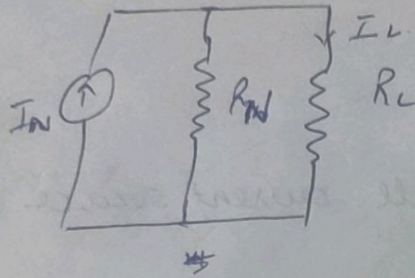
$$R_{Th} = \frac{160}{9} \Omega$$



$$I = \frac{\frac{20}{9}}{\frac{160}{9} + \frac{450}{9}} = \frac{20}{610} = \frac{2}{61} \text{ A}$$

* NORTON'S THEOREM

Acc to Norton's theorem, 'Any ^{linear} bilateral network irrespective of its complexities can be reduced into Norton's equivalent circuit having norton's short circuit ' I_N ' in parallel with Norton's equivalent resistance ' R_N ' in parallel with load resistor R_L



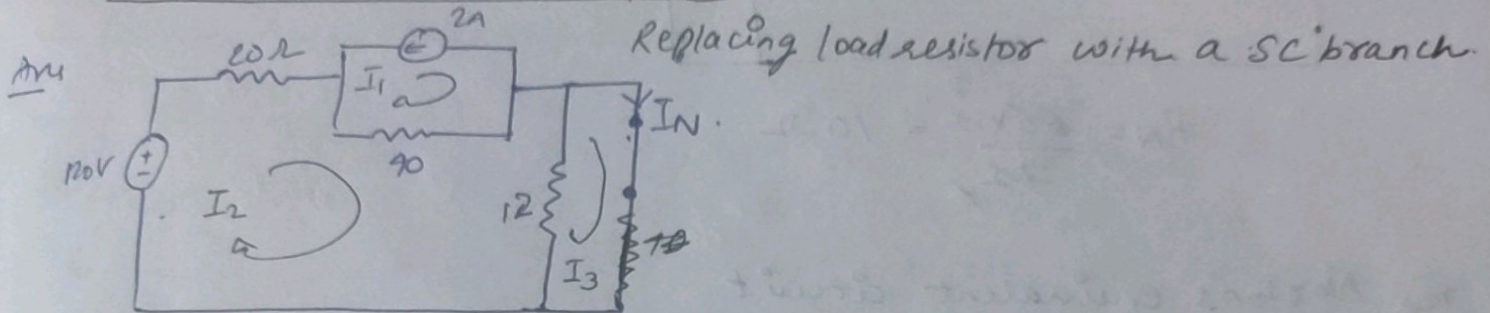
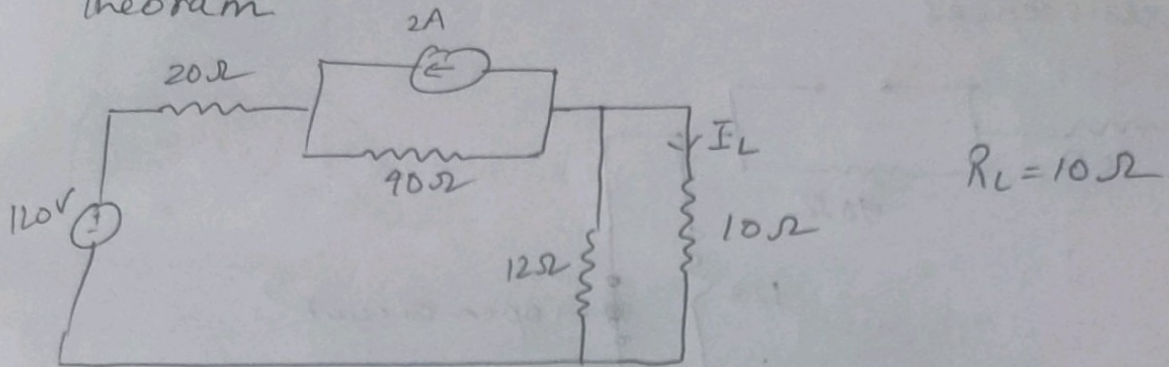
$$I_L = \frac{I_N R_N}{R_N + R_L}$$

* Steps for solving Norton's theorem Numerical:-

- (I): Identify load resistor (R_L)
- (II): Replace R_L with short circuit branch
- (III): The current flowing through this short circuit branch will be norton's current I_N .
- (IV): Remove R_L and replace all active sources by their internal resistance.
- (V): Equivalent resistance across the the two open ends will be Norton's resistance R_N .
- (VI): Draw Norton's equivalent circuit
- (VII): Calculate I_L using Identity

$$I_L = \frac{I_N R_N}{R_N + R_L}$$

Q1 Find the current I in the given circuit having Norton's Theorem



$$I_1 = 2A$$

Applying KVL in Mesh 2

~~$$+120 - 20I_2 - 40(I_2 - I_1) - 12(I_2 - I_3) = 0$$~~

$$+120 - 20I_2 - 40(I_2 - I_1) - 12(I_2 - I_3) = 0$$

$$+120 - 72I_2 + 40I_1 + 12I_3 = 0$$

$$30 - 18I_2 + 10I_1 + 3I_3 = 0$$

$$18I_2 - 10I_1 - 3I_3 = 30 \quad \text{--- (1)}$$

Applying KVL in Mesh 3

$$-12I_3 + 12I_2 = 0$$

$$I_2 = I_3$$

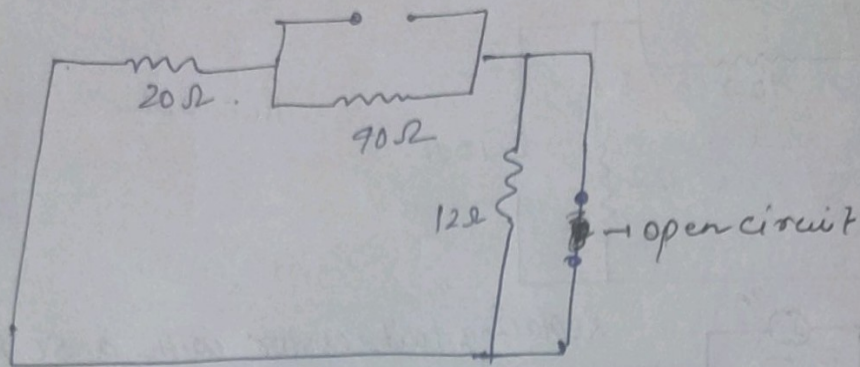
$$18I_2 - 3I_2 + 20 = 30$$

$$15I_2 = 10$$

$$I_2 = \frac{2}{3} = I_3$$

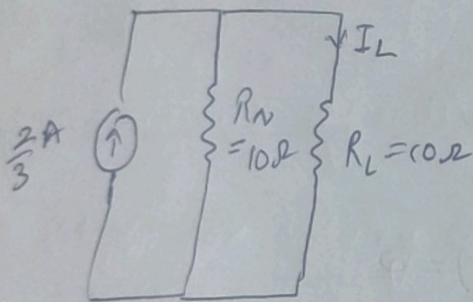
$$I_N = I_2 = \frac{2}{3}A$$

Removing R_L and replacing all the active sources by their internal resistances.



$$R_N = \frac{20 \times 12}{20 + 12} = 7.5 \Omega$$

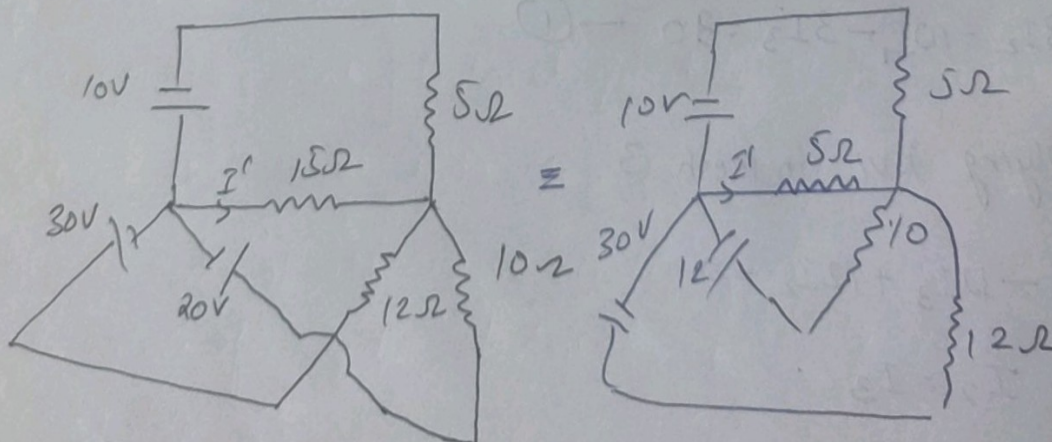
Draw Norton's equivalent circuit



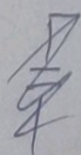
$$I_L = \frac{2/3 \times 10}{20} = \frac{1}{3} A$$

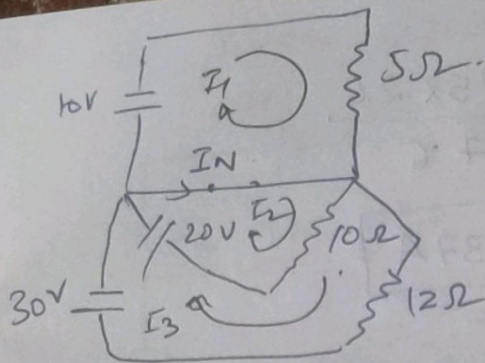
$$I_L = 0.33 A \quad \text{Ans}$$

Q2 Find the current I in given circuit using Norton's theorem



Replacing R_L with SC branch





Applying KVL in Mesh 1

$$-10 - 5I_1 = 0$$

$$\boxed{I_1 = -2A}$$

Applying KVL in Mesh 2

$$-20 - 10I_2 = 0$$

$$\boxed{I_2 = -2A}$$

$$\boxed{I_2 = I_N = 2A}$$

$$-20 - 10(I_2 - I_3) = 0$$

$$-30 - 10I_2 + 10I_3 = 0$$

$$I_3 - I_2 = 2$$

$$I_3 = 2 + I_2$$

$$\boxed{I_3 = -2.5A}$$

$$\boxed{I_N = 4.5A}$$

$$I_N = I_2 - I_1$$

$$I_N = -4.5 + 2$$

$$\boxed{I_N = -2.5A}$$

Applying KVL in Mesh 3

$$-30 + 20 - 10I_3 - 12I_3 + 10I_2 = 0$$

$$-10 - 22I_3 + 10I_2 = 0$$

$$10I_2 - 22I_3 = 10$$

$$-20 - 10 = 22I_3$$

$$I_3 = -1.5A$$

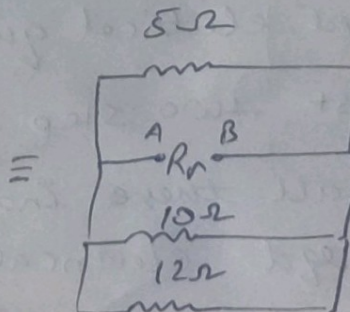
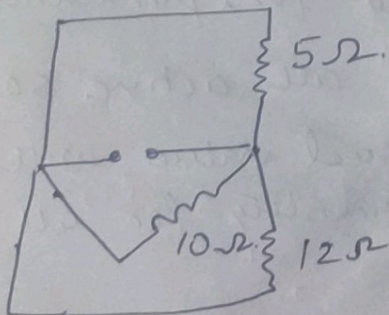
$$10I_2 - 22(2 + I_2) = 10$$

$$10I_2 - 44 - 22I_2 = 10$$

$$-11I_2 = 54$$

$$\boxed{I_2 = -4.5A}$$

For R_N , removing load resistance & replacing all active sources

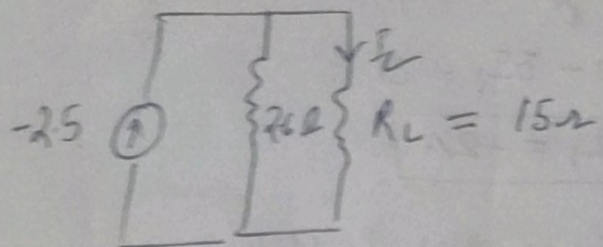


$$\frac{2}{5} \mid \frac{12}{6}, \frac{5}{5}$$

$$R_N = \frac{60}{23} \Omega$$

$$\frac{1}{R_N} = \frac{1}{5} + \frac{1}{10} + \frac{1}{12}$$

$$= \frac{12 + 6 + 5}{60} = \frac{23}{60}$$



$$I_L = \frac{-25 \times 20}{17.6}$$

$$I_L = -0.37A$$

SUPERPOSITION THEOREM

multisource

→ Acc to SPT, in any linear, bilateral, network, the current or voltage across any branch can be calculated by taking algebraic sum of values calculated by taking one source at a time and replacing the other active sources by internal resistances.

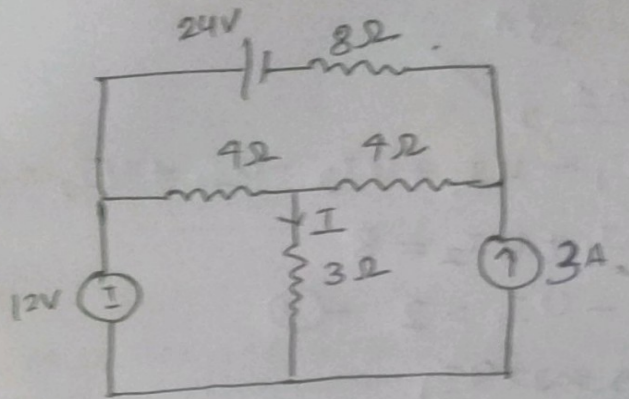
* Why SPT is important?

→ because in SPT theorem, it gives or tell the energy power in circuit by individual active sources

* Steps for solving SPT theorem:-

1. Identify the branch and quantity to be calculated along with presence of more than 1 active sources.
2. Consider any 1 active sources and replace the remaining by internal resistances
3. Calculate the reqd electrical quantity, for ^{that} particular source
4. Repeat the last two step for all active sources
5. Algebraic sum of all these individual value will be final value of reqd electrical quantity for all sources working together.

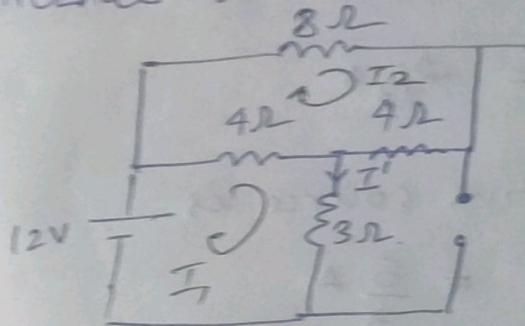
Q1 For given circuit, using the SPT, Calculate I.



There are 3 active sources in given network

Case I: 12V

Taking 12V source & replacing all other active sources by their internal resistance.



Applying KVL in Mesh 1,

$$-4I_2 - 4I_2 + 4I_1 - 8I_2 = 0$$

$$-4I_1 - 16I_2 = 0$$

$$I_1 = 4I_2 \quad \text{--- (1)}$$

Applying KVL in Mesh 2

$$+12 - 4I_1 + 4I_2 - 3I_1 = 0$$

$$12 - 7I_1 + 4I_2 = 0$$

$$12 - 28I_2 + 4I_2 = 0$$

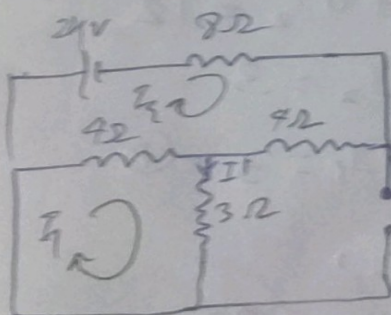
$$24I_2 = 12$$

$$I_2 = 1/2 \text{ A}$$

$$I_1 = 2 \text{ A}$$

$$I' = I_1 = 2 \text{ A}$$

Taking 24V source & replacing all other active sources by their internal resistance



Mesh 1,

$$-4I_1 + 4I_2 - 3I_1 = 0$$

$$7I_1 = 4I_2 \quad \text{--- (1)}$$

Mesh 2,

$$-24 - 8I_2 - 4I_2 - 4I_2 + 4I_1 = 0$$

$$-24 - 16I_2 + 4I_1 = 0$$

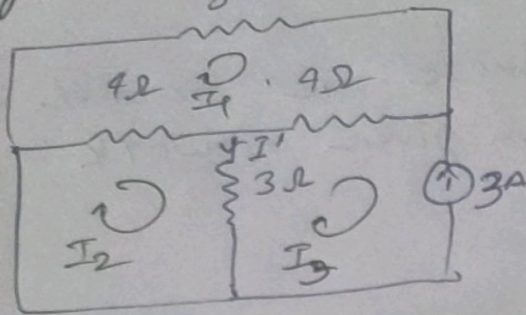
$$-28I_1 + 4I_1 = 24$$

$$I_1 = -1 \text{ A}$$

$$I_2 = -7/4 \text{ A}$$

$$I' = -1 \text{ A}$$

Taking 2A as source



Mesh 1,

$$-4I_2 + 4I_1 - 3I_2 + 3I_3 = 0$$

$$-7I_2 + 4I_1 = 6$$

$$-6I_2 = 8$$

$$I_2 = -4/3 A$$

$$I' = I_2 - I_3$$

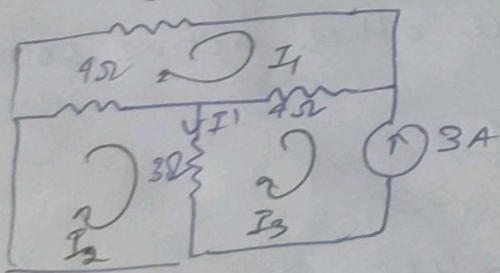
$$I' = -4/3 + 2 = 2/3 A$$

so Algebraic sum of I' for all active source is

$$\frac{2}{3} + 2 - 1 = \frac{2}{3} + 1 = \frac{5}{3} A$$

$$I'_{net} = 5/3 A$$

Taking 3A as source.



$$I_3 = -3A \quad I_1 = -5/4 A$$

$$I' = -2A + 3A = I_2 - I_3$$

$$I' = 1A$$

Mesh 1,

$$-8I_1 - 4I_1 + 4I_2 - 4I_1 + 4I_3 = 0$$

$$-16I_1 + 4I_2 = 12$$

$$-4I_1 + I_2 = 3$$

$$I_2 - 4I_1 = 3 \quad \text{--- (1)}$$

Mesh 2,

$$-4I_2 + 4I_1 - 3I_2 + 3I_3 = 0$$

$$-7I_2 + 4I_1 = 9$$

$$-7I_2 + I_2 = 12$$

$$-6I_2 = 12 \quad I_2 = -2A$$

So Algebraic Sum is $1A + 2A - 1A$

$$\boxed{I'_{net} = 2A}$$

$$I'_{net} = I'_1 + I'_2 + I'_3$$

$$\boxed{I'_{net} = 2A}$$

MAXIMUM POWER TRANSFER THEOREM

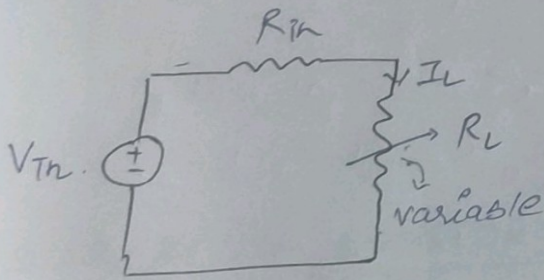
→ Acc to maximum power ~~energy~~ transfer theorem, the condⁿ for maximum power flow through load resistor R_L can be achieved when load resistor equals Thevenin equivalent resistance of given circuit.

$$\boxed{R_L = R_{Th}} \text{ --- condⁿ for above theorem}$$

→ It states that amt of ^{maximum} power dissipated through load resistor.

Steps to find Max Power:-

- 1) Same upto converting circuit into thevenin circuit
- 2) Then solve for maximum power.



Power through R_L

$$P = I_L^2 R_L$$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} \text{ (from thevenin circuit)}$$

$$P = \left[\frac{V_{Th}}{(R_{Th} + R_L)} \right]^2 R_L$$

Differentiating above eqⁿ w.r.t R_L and equating it to 0.

$$\frac{dP}{dR_L} = 0$$

$$\frac{dP}{dR_L} = (V_{Th})^2 \left(\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right) = 0$$

$$(R_{Th} + R_L)^2 = 2R_L(R_{Th} + R_L)$$

$$2R_L = R_{Th} + R_L$$

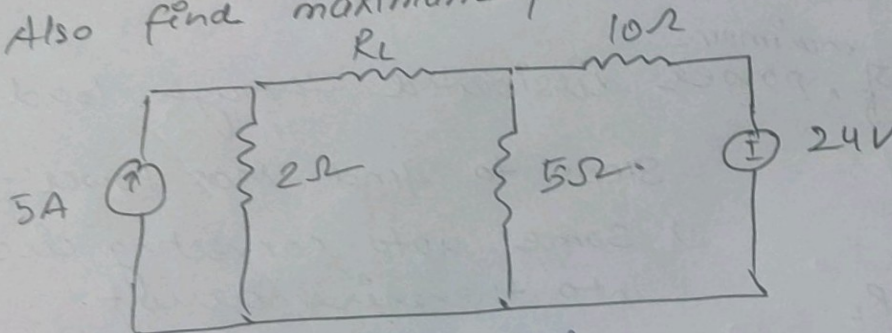
$$\boxed{R_L = R_{Th}} \text{ reqd condn for maximum power flow}$$

(Putting $R_L = R_{Th}$)

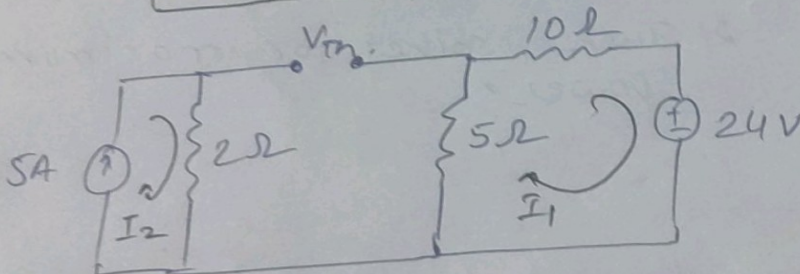
$$P = \left(\frac{V_{Th}}{2R_{Th}} \right)^2 \cdot R_{Th}$$

$$\boxed{P_{\max} = \frac{(V_{Th})^2}{4R_{Th}} = \frac{V_{Th}^2}{4R_L}}$$

Ques In given network find value of R_L which will absorb the maximum power from the source. Also find maximum power.



Ans

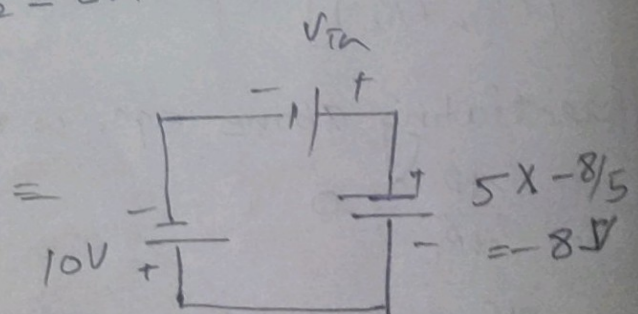
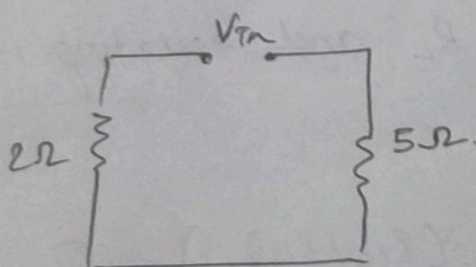


Mesh 1 ,

$$-10I_1 - 5I_1 = 24$$

$$-15I_1 = 24$$

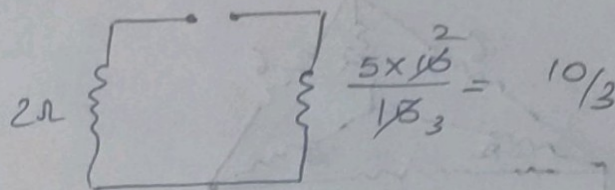
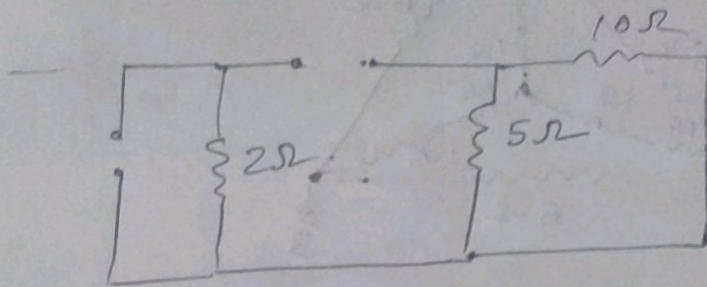
$$I_1 = -8/5 \text{ or } I_2 = 5A$$



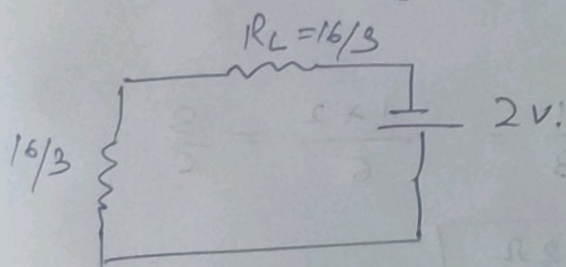
$$V_{Th} + 8 - 10 = 0$$

$$\boxed{V_{Th} = 2V}$$

Replacing all sources by internal impedance.



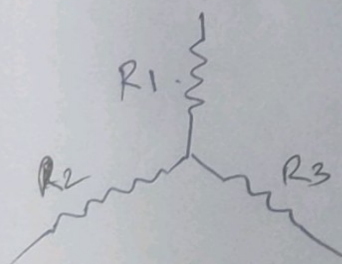
$$R_{Th} = 2 + \frac{10}{3} = \frac{16}{3} \Omega$$



$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{2^2}{4 \times \frac{16}{3}}$$

$$P_{max} = \frac{3}{16} W$$

STAR-DELTA TRANSFORMATION

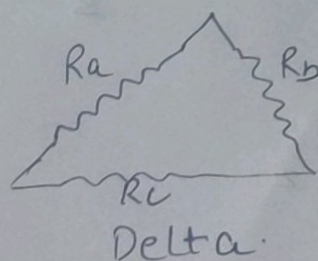


Star
Delta to star

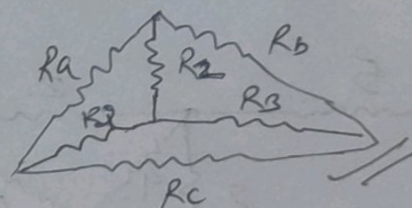
$$R_1 = \frac{R_a \cdot R_b}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a \cdot R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_b \cdot R_c}{R_a + R_b + R_c}$$



Delta.



Star delta conversion

star to delta

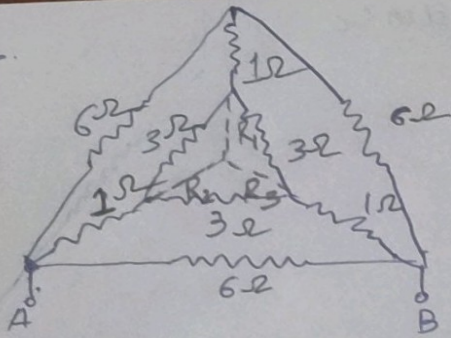
$$R_a = \frac{R_1 + R_2 + \frac{R_1 \cdot R_2}{R_3}}{R_3}$$

$$= \frac{R_3 R_1 + R_2 R_3 + R_1 R_2}{R_3}$$

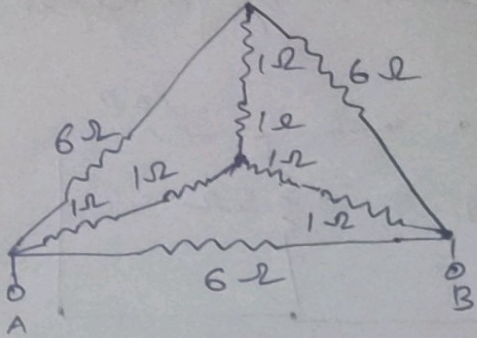
$$R_b = \frac{R_1 R_2 + R_3 R_2 + R_1 \cdot R_3}{R_2}$$

$$R_c = \frac{R_2 R_1 + R_3 R_1 + R_2 \cdot R_3}{R_1}$$

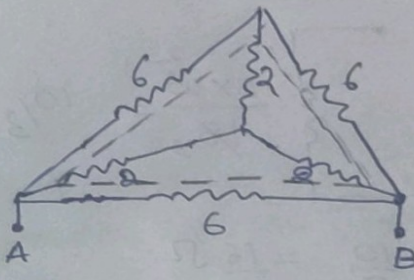
Q1



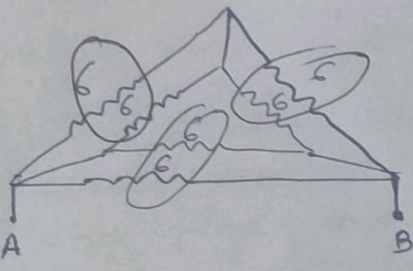
≡



III



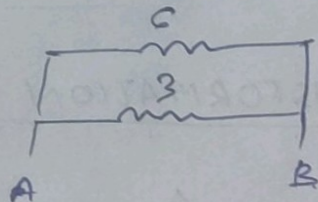
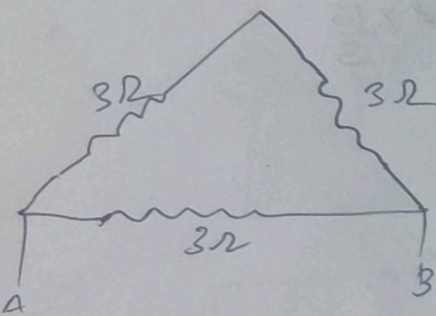
$$R_1 = \frac{3 \times 3}{9} = 1\Omega = R_2 = R_3$$



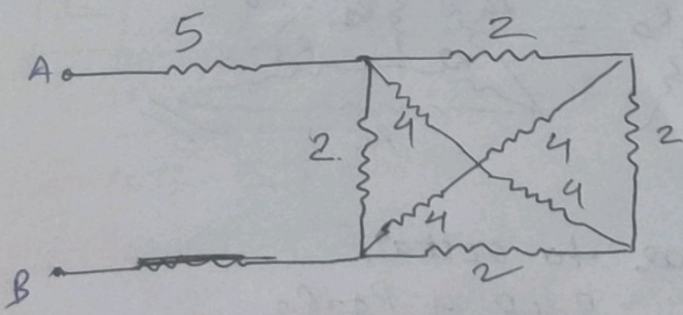
≡

$$\frac{1}{R_{eq}} = \frac{1}{6} + \frac{1}{3} = \frac{1+2}{6} = \frac{3}{6}$$

$$R_{eq} = 2\Omega$$



Q2



Ans